Heat and Moisture Exchanger (HME) *Prodigy Anesthesia*

Heat and moisture exchangers (HME) are inexpensive, silent, easy-to-use devices that help retain heat and moisture within the anesthesia circuit. Advantages of HMEs include: no need for water or electrical power source and no risk of hyperthermia, overhydration, burns, or electrical shock. They all act as large particle filters and many serve as effective bacterial and viral filters.



The primary disadvantages of HMEs are that they are not nearly as effective at warming and humidifying the patient's airway as water-based, electrical devices. They also increase deadspace and can increase the work of breathing.

Airway obstruction can occur if the HME becomes blocked with fluid, blood, secretions, nebulized drugs, or if a mechanical defect is present in the device. If increased airway resistance is experienced during an anesthetic, the peak pressure should be measured both with and without the HME in place. Nebulizers and metered-dose inhalers should be placed between the HME and the patient during treatment.

There are two primary types of HME's: hydrophobic and hygroscopic. Hydrophobic HME's have a hydrophobic membrane with small pores that is pleated to provide a greater surface area. Hygroscopic HME's contain a wool, foam, or paperlike material that is coated with a chemical that helps it to retain moisture.

High ambient temperature may decrease the effectiveness of hydrophobic HME's. Compared to hygroscopic HME's. Hydrophobic HME's may also be more effective at preventing the transmission of the hepatitis C virus than hygroscopic HME's. Also, if hygroscopic HME's become wet, they may lose their ability to filter airborne pathogens and airway resistance may increase substantially. The airway resistance of hydrophobic HME's increases only slightly if wet. Hygroscopic HME's are considered more effective at preserving heat and humidity than hydrophobic HME's.

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Dorsch JA, Dorsch SE. Understanding Anesthesia Equipment. 5th ed. Philadelphia, PA: Lippincott Williams and Wilkins, 2008: 299-301.

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